

ACCURATE PLATE REVERB PARAMETER ESTIMATION USING TWO-STAGE EVOLUTIONARY SEARCH

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ABSTRACT

We describe our submission to Task A of the 1st DAFx parameter estimation challenge. The task is to recover the six physical parameters of a simulated metal-plate reverberator — its dimensions and material properties — from a single impulse response (IR). We treat this as a black-box optimization: candidate parameter sets are fed to the simulator and scored by a loss against the target IR. The method has two stages. The first uses CMA-ES, an evolutionary optimizer, to recover five of the six parameters, comparing IRs under an amplitude-normalized loss. Amplitude normalization makes the search robust but discards the cue to the sixth parameter, the plate’s surface density; a second stage therefore estimates it alone, with a ternary search on the un-normalized loss. As the choice of loss strongly affects the search, we select it beforehand by comparing a range of candidate losses.

1. INTRODUCTION

A plate reverberator [1] produces artificial reverberation [2] by picking up the vibration of a driven metal plate; such units are now commonly emulated in software. Given a simulated plate IR, one may wish to recover the physical parameters that generated it, both to learn useful configurations and as a step towards broader creative applications.

The 1st DAFx parameter estimation challenge [3] benchmarks this problem. It consists of two tasks: Task A, “physical parameter identification”, is to infer the physical parameters of the plate from a simulated IR, while Task B, “modal parameter estimation”, is to dissect the modes of vibration. In this work, we describe our submission to Task A.

Existing methods do not transfer readily. Xu and Ding [4] adapt the Newton method to estimate coefficients of linear time-invariant systems, but it has been demonstrated only on simple third-order systems and does not readily scale to a problem of this size. Lee *et al.* [5] optimize a differentiable room reverberator via gradient descent; our preliminary attempts to adapt this approach were unstable and prone to divergence, likely reflecting a rugged, ill-conditioned loss landscape. We therefore adopt the covariance matrix adaptation evolution strategy (CMA-ES) [6], a derivative-free optimizer suited to exactly such problems. CMA-ES has also

shown promise in similar parameter-estimation settings, such as articulatory speech synthesis [7].

2. BACKGROUND

2.1. Plate Reverb Simulator, and Task A of the Challenge

The plate simulator [3] implements the general solution of the damped Kirchhoff–Love equation [1, 8], governed by the set of physical parameters

$$P := \{\rho, E, \nu, L_x, L_y, h, T_0, \eta_0, \eta_1, x_i, y_i, x_o, y_o\}, \quad (1)$$

which are, in order: the volumetric density (ρ), Young’s modulus (E), Poisson’s ratio (ν), the width (L_x), the height (L_y), the thickness (h), the tension applied via connections to the chassis (T_0), decay constants (η_0, η_1), the input location (x_i, y_i), and the output location (x_o, y_o). We refer the reader to the challenge documentation [3] for the full model. For our purposes, a relevant property is that the simulator output is a sum of second-order infinite impulse response (IIR) filters whose coefficients are functions of the physical parameters. We therefore estimate the parameters by optimizing P directly, searching for the parameter set whose rendered IR best matches the target.

Task A of the challenge is to estimate $P_{\text{true}} := \{x_o, y_o, L_y, \mu, T_0/\mu, D/\mu\}$, a derived subset of P associated with a given IR, where $\mu := \rho h$ is the surface density and $D := Eh^3/12(1 - \nu^2)$ is the flexural rigidity. Other parameters ($\nu, L_x, \eta_0, \eta_1, x_i, y_i$) remain fixed. This task design allows for a one-to-one correspondence between the IR and the six parameters, and thus a fair evaluation via normalized mean square error (NMSE):

$$\text{NMSE}(P_{\text{true}}, P_{\text{est}}) = \frac{1}{6} \sum_{i=1}^6 \left(\frac{P_{\text{true},i} - P_{\text{est},i}}{M_i - m_i} \right)^2 \quad (2)$$

where M_i and m_i are the upper and lower bounds of the values each parameter may assume, while P_{true} and P_{est} are the 6-dimensional vectors of true and estimated parameters. The bounds are derived from the legal ranges of the raw quantities $E, \rho, h, L_y, T_0, x_o$, and y_o .

2.2. Covariance Matrix Adaptation Evolution Strategy

Covariance matrix adaptation evolution strategy (CMA-ES) [6] is a derivative-free optimizer for continuous black-box objectives. At each generation, it samples a population of candidate solutions from a multivariate Gaussian, evaluates them, and adapts the distribution: the mean is moved towards the rank-weighted best candidates, while the covariance matrix and a global step size are up-

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dated to align the search with the directions in which good candidates have consistently been found [6].

CMA-ES is designed for ill-conditioned, non-separable problems with many local optima (i.e. “multimodal”) [6]. Its covariance matrix learns the local geometry of the objective, providing implicit preconditioning of the search space, while its rank-based selection yields invariance under monotonic transformations of the objective. These properties suit the present problem: the inverse mapping from IR to parameters is also expected to be ill-conditioned and multimodal, the regime CMA-ES is built for. We note that, although the simulator is at least piecewise differentiable, direct gradient descent on it tended to diverge, and proved difficult to stabilize through hyperparameter tuning.

3. METHODOLOGY

Our method estimates the six Task A parameters in two stages: a CMA-ES search over the seven raw physical parameters, followed by a dedicated refinement of the mass density μ . CMA-ES searches the raw parameters rather than the six Task A composites directly: the raw parameters are the simulator’s native inputs, each with an independent range, so the search domain is a simple box. The composites are products and ratios of these (Section 2.1) and lack independent bounds, so searching them would require handling the constraints coupling them. The six evaluation parameters are then computed from the estimated raw parameters.

Both stages minimize a common objective, an L_1 distance on STFT magnitudes between the target IR and one rendered by the plate simulator. We selected this loss by comparing several spectral-magnitude, perceptual-filterbank, phase-aware, and time-domain losses, including spectral-convergence loss [9], L1 loss on log-mel spectrogram, L1 loss on group-delay, and L1 loss on constant-Q transform (CQT), retaining the one that converged most reliably and yielded the lowest mean NMSE on a set of generated IRs. All IRs are processed at 44.1 kHz and truncated to the first 0.25 s. The STFT uses a 4096-point Hann window with a hop of 1024 and the centered, reflection-padded framing of the standard transform. A PyTorch port of the plate simulator is used to leverage GPU parallelism; additionally, we derive and use the analytical response of the second-order IIRs to further parallelize the plate.

3.1. Stage 1: CMA-ES Search

We found CMA-ES searches over a seven-dimensional space of E , ρ , h , L_y , T_0 , x_o , and y_o to recover all parameters well except μ . A single run updates a population of seven-element vectors until one of four conditions is met: a budget of 30,000 objective evaluations, a tolerance of 1×10^{-5} defined on the range of recent best losses values, or a per-generation or per-trial wall-clock limit (300 and 600 seconds respectively). Each candidate is rendered by the simulator and compared to the target. As IR amplitudes can vary widely, IRs are peak-normalized before a loss value can be derived. This makes the loss comparable across IRs and removes some failure modes.

Each restart draws its initial distribution mean by Latin hypercube sampling (LHS) in a normalized $[-1, 1]^7$ cube, for diversity across restarts; CMA-ES then operates throughout in this normalized space with initial step size $\sigma_0 = 0.6$, mapping affinely onto the physical bounds of E , ρ , h , L_y , T_0 , x_o , y_o to render a candidate. Optuna [10] prunes unpromising runs via a SuccessiveHalvingPruner (reduction factor 3, min resource 15) and selects each

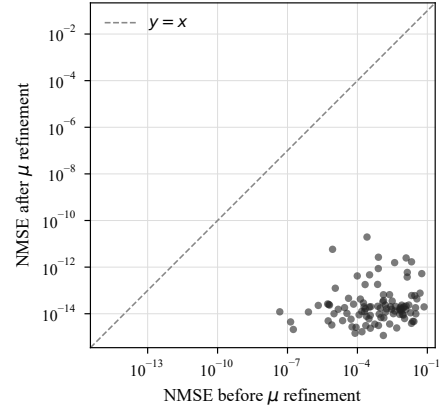


Figure 1: Per-IR NMSE (log-log) before and after Stage 2 μ -refinement, over 100 generated IRs. All points fall well below $y = x$, with refined NMSE often on the order of 10^{-14} or lower.

run’s population size in [30, 60], as large populations were sometimes detrimental. As the ground truth is unavailable at inference time, restarts are not governed by parameter error: they halt once any completed run reaches a loss below 0.01, an empirically chosen threshold, or after 400 restarts.

3.2. Stage 2: Surface-Density Refinement

The peak normalization employed in stage 1 removes the absolute amplitude, and with it the only cue to μ : with the ratios D/μ and T_0/μ and the plate geometry held fixed, μ is monotonic to the overall amplitude of the response. Stage 1 therefore recovers μ poorly. We address this with a separate 50-iteration ternary search over μ , holding the other five parameters at their Stage 1 estimates and minimizing the same objective without normalization. The search spans $\mu \in [2.43, 106.15]$, the range induced by the bounds on ρ and h .

4. METHOD VALIDATION

To verify the two-stage design, we ran the full pipeline on 100 IRs generated from the plate simulator with known parameters. Figure 1 reports the NMSE of each IR before and after the Stage 2 ternary search. All 100 IRs improve: the mean NMSE falls from 6.7×10^{-3} after Stage 1 alone to 3.9×10^{-13} after refinement (median 1.6×10^{-14}), an improvement of roughly ten orders of magnitude. Stage 1 takes a median of 36 s per IR (mean 88 s), with most of the total runtime concentrated in a small number of harder IRs that required more CMA-ES restarts. No IR, however, exceeded our 400-restart cap, with the maximum at 233 restarts. Stage 2 then contributes the NMSE improvement above at negligible cost — approximately 0.5 s per IR — validating the two-stage architecture as a substantial accuracy gain for minimal additional runtime. Because Stage 2 adjusts only μ , this also confirms that Stage 1 had already recovered the other five parameters (D/μ , T_0/μ , L_y , x_o , y_o) to comparable precision.

5. CONCLUSION

We have described our submission to Task A of the 1st DAFx parameter estimation challenge, in which an IR from a plate simulator is inverted back to its six generating physical parameters. We adopted a two-stage optimization approach: a CMA-ES search that recovers five of the six target parameters (D/μ , T_0/μ , L_y , x_o , y_o), followed by a ternary search that refines the surface density μ . On our validation set of 100 IRs, our method reaches NMSE values of $\sim 10^{-14}$ in about two minutes per IR. Our solution has been run on the challenge’s evaluation set; the results will be determined by the organizers.

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